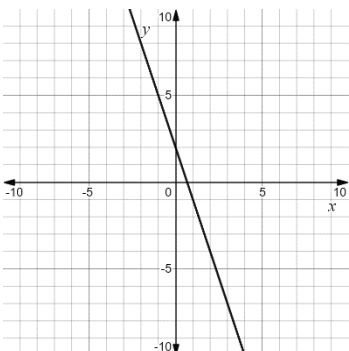


Algebra 1
Unit 3
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

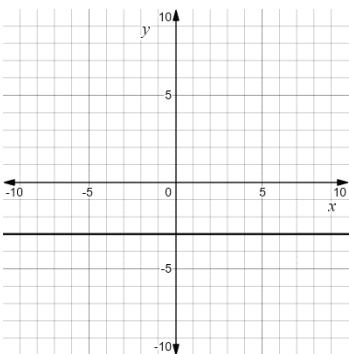
For questions 1 through 4, determine the slope of the line shown.

1.



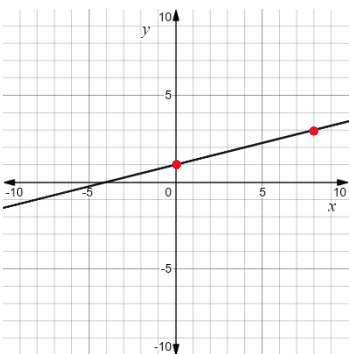
$$m = \underline{\quad -3 \quad}$$

2.



$$m = \underline{\quad 0 \quad}$$

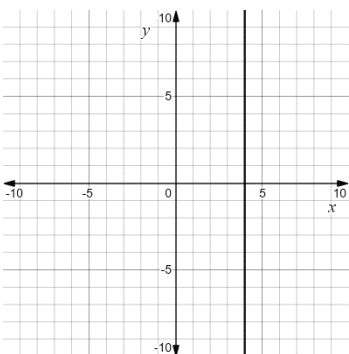
3.



$$\begin{aligned} (x_1, y_1) &= (0, 1) \\ (x_2, y_2) &= (8, 3) \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{3 - 1}{8 - 0} = \frac{2}{8} \\ &= \frac{1}{4} \end{aligned}$$

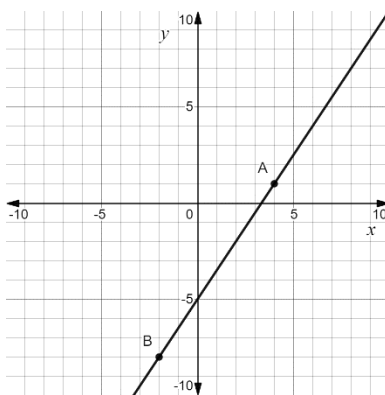
$$m = \underline{\quad \frac{1}{4} \quad}$$

4.



$$m = \underline{\quad \text{undefined} \quad}$$

For questions 5 through 8, the graph of a line with points A and B labeled is shown.



5. Write the equation of the line in point-slope form using point A .

$$m = \frac{3}{2} \quad (4, 1) \quad y - 1 = \frac{3}{2}(x - 4)$$

6. Write the equation of the line in point-slope form using point B .

$$m = \frac{3}{2} \quad (-2, -8) \quad y + 8 = \frac{3}{2}(x + 2)$$

7. Rewrite the equation in point-slope form from point A in slope-intercept form.

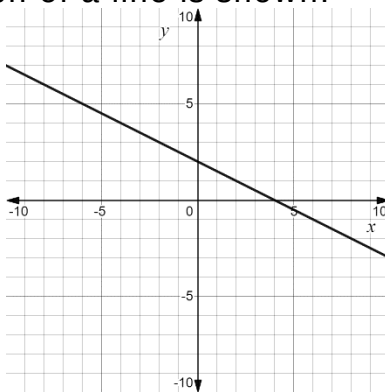
$$y - 1 = \frac{3}{2}x - \frac{12}{2} \rightarrow y - 1 + 1 = \frac{3}{2}x - 6 + 1 \quad y = \frac{3}{2}x - 5$$

8. Rewrite the equation in point-slope form from point B in standard form.

$$y + 8 = \frac{3}{2}x + 3 \rightarrow y - \frac{3}{2}x = 3 - 8$$

$$-\frac{3}{2}x + y = -5 \text{ or } -3x + 2y = -10$$

For questions 9 and 10, the graph of a line is shown.



9. Write the equation of the line in point-slope form.

$$m = -\frac{1}{2} \quad (0, 2) \quad y - 2 = -\frac{1}{2}(x - 0)$$

10. Rewrite the equation from question 9 in standard form.

$$y - 2 = -\frac{1}{2}x \rightarrow y - 2 + 2 = -\frac{1}{2}x + 2 \rightarrow y + \frac{1}{2}x = -\frac{1}{2}x + 2 + \frac{1}{2}x$$

$$2(y + \frac{1}{2}x = 2) \rightarrow x + 2y = 4$$